SIADS 696 Milestone II Project Report
Identifying Data Centers Based on Energy Usage, Location, and Structural Characteristics
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Introduction

Throughout communities across the United States, there is a growing concern due the rapid development of data centers, driven in part by the increasing demand for artificial intelligence. As the AI industry continues to expand, some of the existing data centers, originally built for operations such as crypto mining, are beginning to transition their operations to the AI sector for more profitability (Tidy, 2026). This wave of transitions paired with the continued expansion of data centers, could impact the demand for utilities such as water and electricity, potentially affecting local residents.

With this in mind, we began our project by asking: Are there meaningful differences in facility characteristics that help distinguish between different types of data centers? Understanding the differences between facilities is an important step toward evaluating their potential impact on local infrastructure. Currently, many open source data resources lack clear classifications of data center purpose, making it difficult to differentiate between facility types. Our project seeks to uncover patterns in data center features and use these differences to classify facilities into categories related to their operational purpose.

Our unsupervised approach is aimed at determining whether distinct archetypes of data centers naturally emerge without providing any information related to the type or use of the data center. We employed KMeans, Gaussian Mixture Model (GMM), and Agglomerative clustering models to investigate the individual types of data centers that exist and the attributes that differentiate them. The final, fine-tuned Gaussian Mixture Model provides the best overall performance and identifies three general data center archetypes: large facilities with a limited number of nearby generators and lower electricity cost, medium-sized facilities with substantial surrounding energy infrastructure, and small facilities with limited capacity but high regional electricity consumption.

In the supervised part, we focused on cryptocurrency mining, and asked whether public facility characteristics can identify them. We built a crypto label by matching FracTracker's data center list to its crypto mining tracker and to facilities that describe themselves as crypto. We then compared logistic regression, a random forest and k-nearest neighbors on all 1,720 facilities.

The tuned random forest was the best classifier, with a cross-validated F1 of 0.32, with size being the most important group of features. Crypto labels are rare (47 of 1,720), which makes the task difficult, but the forest's precision on held-out data was about six times the base rate. At least one of the model failures, the Lake Mariner facility, is a known crypto site that our label missed and that is now converting to AI.

Related Work

1. **Lawrence Berkeley National Laboratory's 2024 United States Data Center Energy Usage Report**

This report examines historical electricity consumption across US based data centers and develops projections of future energy demand through 2028. These estimates are largely based on the estimated number of servers installed in the facilities, and they exclude other forms of data centers, like cryptocurrency mining, from the main estimate. Unlike this research, our project focuses on identifying if differences in energy usage among individual data centers help label data centers when the type is not reported. This project is an inverted look at similar datasets, where the report starts with equipment and predicts energy usage, we are starting with energy usage and predicting the equipment and purpose. (Shehabi et al., 2024)

2. **Tracking electricity consumption from U.S. cryptocurrency mining operations**

   In this study, the EIA analyzes individual cryptocurrency mining operations using their electricity consumption and usage reports taken on a monthly basis. Although this study considers individual facilities, it primarily focuses on a single type of facility, cryptocurrency, and will not be reflective of electrical consumption once those facilities are upgraded for AI purposes. Our project attempts to broaden this study's findings, examining possible differences in characteristics across facility types, to aid in the understanding of their individual impact on electricity demand. One of the goals is to allow electrical grid providers to prepare and evaluate the impact on their entire grid when facilities are migrated from cryptocurrency to AI. (Morey et al., 2024)

3. **Characteristics and Energy Use of Volume Servers in the U.S.**

   This report investigates the hardware characteristics of the servers themselves rather than the overall facility, testing server configurations and comparing their idle energy consumption to benchmarks set by Energy Star and SPEC servers. While this study takes a more in depth look at the individual servers, it primarily focuses on idle usage which normally accounts for 50-70% of full-load power and may leave out important details related to energy usage during peak data center's operation. Our project expands on this study, examining data center on a macro scale to determine whether facility characteristics potentially reveal any significant differences in energy usage or operational type regardless of the number of servers and removing assumptions about idle run time. (Fuchs et al., 2018)

Data Sources

Our analysis covered a wide variety of data sources to ensure we gained as much information regarding the primary list of data centers as possible. These resources were retrieved from multiple open source, community-based datasets and merged together to form a final comprehensive dataset for both our supervised and unsupervised models.

Dataset	Information	Location
FracTracker U.S. Data Centers	- Open source, downloadable CSV file - 1,720 facilities/records	Link

	- 53 fields including (name, operator, location, status, purpose, size rank (mw), facility size, building sqft, project cost) - Updated daily, downloaded 09/2026	
FracTracker Cryptocurrency Mining Operations	- Open source, downloadable CSV file - 302 crypto-mining operations - 18 fields including (name, company, location, power source, mw, status) - Last updated 09/2022	Link
Compute Atlas Data Centers	- Open source, downloadable CSV file - 2,659 facilities/records - 20 fields including (name, operator, location, facility type, operational / planned mw capacity, status) - Continuously updated, downloaded 10/2026	Link
EIA-861M Retail Sales of Electricity Data	- Open source, downloadable XLSX file - 10,150 records - 24 fields including (commercial customer count, price per kWh, consumed MWh, state, year) - Updated annually (2010-2025), parsed down to use 2025 dataset	Link
EIA-860 Generator Data	- Open source, downloadable XLSX file - 27,919 generator records - 73 fields including (state, county, summer and winter MW capacity, time from start up to full load in minutes, energy source type, total generator capacity) - Updated annually, 2025 dataset used	Link

Table #1 - Data Sources

Feature Engineering

Major steps in feature engineering:

1. Join the FracTracker data centers with the FracTracker crypto tracker and Compute Atlas
    a. Standardize address syntax (regex matching and removing whitespace)
    b. Match crypto tracker sites by exact address or within 0.075 mi (29 matches)
    c. Match Compute Atlas facilities within 0.075 mi (197 matches) to fill in missing values
    d. Fill missing values for mw, capacity, and power source
    e. Standardize units and split any ranges into min/max columns
    f. Normalize power source naming syntax
    g. Create binary column if row meets “crypto” requirement
    h. *Note: even after joining these two datasets there was a high rate of missing data, instead of filling that categorical data in, we opted to using missing data as a feature in our models
2. EIA-860 Generator Data
    a. Copy helpful features into new dataframe

    b. Populate energy source code with full energy source description from EIA provided source codes sheet
    c. Rename columns for clarity
    d. Standardize time values from various strings to numerical value of minutes
    e. One-hot encoding for energy source per generator
    f. Split numerical values between summer/winter seasons as generator capacity and performed varies seasonally - size(count),max, sum, and mean values used depending on column data
    g. Add column for total generator capacity across all seasons
3. EIA-860M
    a. Convert year to type string
    b. Create a copy to dataframe that only includes 2025 and commercial customers as most states zone data centers as commercial partners
    c. Assign season based on reporting month
    d. Group by state and season to get an overall view of energy cost and consumption of commercial customers
    e. Pivot dataset so each state has a singular row that denotes summer/winter/total
4. Joined Dataset for Modeling
    a. Join generator data to data center dataset based on county
    b. Join energy sales and consumption data to data centers based on state
5. Please see Appendix for the Final Feature Table

Unsupervised Learning

Motivation

The goal of this unsupervised learning task was to determine whether distinct archetypes of data centers emerge naturally without providing any information related to the type or use of the data center. Using a combination of features from the FracTacker and EIA datasets, primarily related to facility size, capacity, and surrounding energy infrastructure, we would use a variety of clustering algorithms to discover any meaningful relationships. These results would help us further investigate the individual types of data centers that exist and the attributes that differentiate them.

Data Preprocessing

Prior to model development, we conducted a data preparation phase that involved analyzing feature availability, removing observations with missing information, and transforming and scaling features. Missing data presented significant limitations to our analysis and modeling approach, as several potentially informative attributes such as cooling systems, power sources, and project costs were sparsely populated. Although imputation was considered, the number of rows containing valid observations made up less than 20% of the dataset. Imputing such a large portion of missing values may have introduced artificial relationships that reflected the missingness rather than relationships associated with the data centers themselves. Therefore, we prioritized 256 data centers with high feature availability to ensure the unsupervised models reflected valid, interpretable relationships.

The final features selected included seven variables: *'sizerank_numeric', 'facility_size_sqft', 'property_size_acres', 'generator_count', 'capacity_sum', 'summer_Cents/kw.1', and 'summer_Megawattshours.1'*. These features not only represent a data center's physical attributes such as size, acres, and MW capacity but also information relevant to the energy infrastructure that surrounds

them. All variables within this study were numerically represented, aside from size rank which is a categorical representation for a capacity range. Therefore, log transformations and scaling have been applied to the selected features in order to reduce the disproportional influence of large observations.

Methods

Once features were chosen, three clustering techniques were used to explore different mechanisms for identifying groups within the dataset: K-Means, Gaussian Mixture Models (GMMs), and Agglomerative Hierarchical Clustering. After fitting each model on the scaled data, we used a range of values (2-7) in order to examine how changing the number of clusters impacted cluster quality.

K-Means clustering was first applied to provide a baseline for identifying groups that may form based on similar data center characteristics. This method uses a simple, interpretable approach, assigning each datapoint to the cluster with the nearest centroid.

Gaussian Mixture Models were selected to further investigate data center groupings and whether more flexible cluster boundaries could better represent the relationship within our dataset. While tuning the model, the covariance structure 'tied' appeared to better capture the correlation between size related variables in our select features, improving the overall performance.

Agglomerative clustering was used as an alternative clustering technique that could capture more hierarchical tendencies in our data set. We experimented with a variety of linkage techniques, however, there were no significant performance gains.

Evaluation

Due to the limited number of true archetype labels within our dataset and with the interest of discovering new, unidentified groups, we utilized metrics which evaluated the quality of the clusters created by each model. Silhouette score was used to measure how well individual observations fit within their assigned cluster and Davies-Bouldin was used to evaluate within-cluster distances relative to the distance between cluster centroids. Rather than using these metrics on a single cluster size, we performed hyperparameter tuning using a range of values from 2 to 7 to directly compare cluster quality between each model.

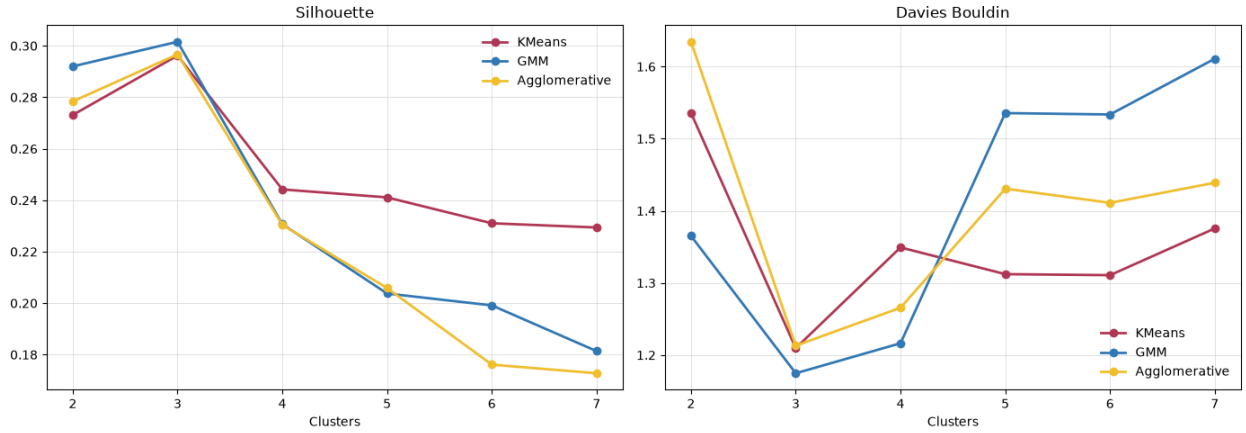


Figure #2 - Unsupervised Model Comparison

Both of the evaluation metrics indicate that a threshold of three clusters represents the strongest clustering configuration across each of the algorithms. Specifically, GMM shows the strongest performance of all models achieving the highest Silhouette score of 0.30 and the lowest Davies-Bouldin index of 1.17.

K-Means and Agglomerative models show slightly weaker performance overall with peaks and valleys in the same areas as GMM.

Additionally, Principle Component Analysis (PCA) was applied to project the features onto a two-dimensional space to observe the clusters created by each model.

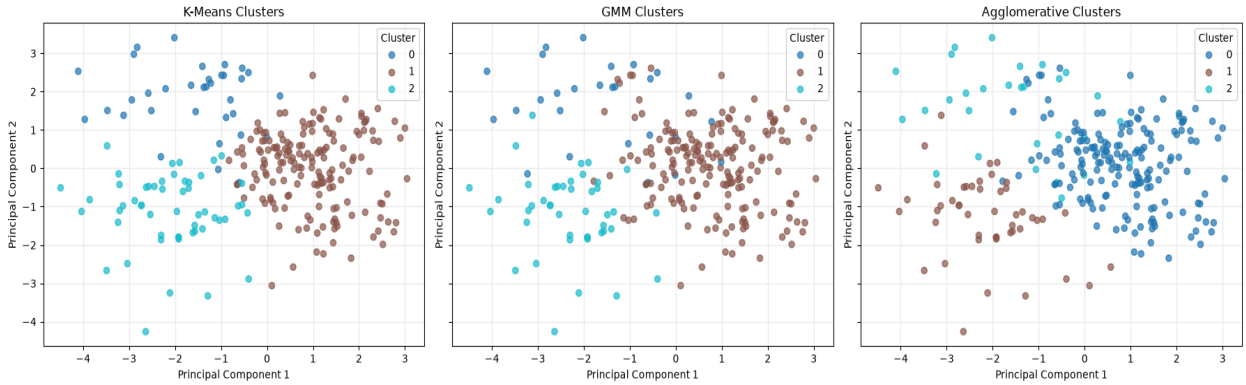


Figure #3 - PCA Clusters for Finetuned Unsupervised Models

The resulting clusters from the top performing model (GMM) indicate three general subgroups of data centers. Large facilities with fewer nearby generators and lower electricity cost (cluster 1), medium-sized facilities with substantial surrounding energy infrastructure (cluster 0), and small facilities with limited capacity but high regional electricity consumption (cluster 2). One possible explanation for this insight is that larger facilities may be built in remote locations where land is more affordable, whereas smaller to mid-size facilities are built in more urban locations where land is more expensive. This could provide some insight into why there is less energy infrastructure and lower energy costs associated with the larger data centers versus the small to mid-sized facilities.

Sensitivity Analysis

To evaluate the robustness of our best-performing model, GMM, we removed singular features related to the surrounding energy infrastructure, including *'generator_count', 'capacity_sum', 'summer_Cents/kw.1',* and *'summer_Megawattshours.1',* clustering data centers on three of the four infrastructure attributes at a time. Overall performance metrics remained relatively the same when removing the features with small shifts in Silhouette and Davies-Boudlin around 0.05 units. However, there was a shift in the optimal clusters (3 → 2) when removing energy cost and the number of nearby generators indicating that these features may be an influential factor in distinguishing between data centers.

Supervised Learning

Our goal was to see whether public facility characteristics can identify crypto-mining data centers. FracTracker's data center list does not label crypto facilities, so we built a label from two sources. We first matched each data center to FracTracker's crypto mining tracker, either by exact address (20 matches) or by location within 0.075 miles (9 matches). We then flagged data centers whose own purpose or name mentions crypto or bitcoin (23). After removing 5 that appeared in both, 47 of the 1,720 data centers

(2.7%) are labeled crypto. We held out 25% as a test set, stratified so both sets keep the same share of crypto sites, leaving 1,290 training rows (35 crypto) and 430 test rows (12 crypto).

We built our features around seven groups of features about each data center: its operating status, how big it is, which fields FracTracker left blank, how it is powered and cooled, whether the community has pushed back, media coverage, and electricity price. Each question became a feature group, and each group then had to be turned into columns a model could read.

Status was the simplest and most fully populated. FracTracker places every site in one of seven stages, from pre-proposal to operating to cancelled, and we one-hot encoded them into 7 columns. Size needed more care. FracTracker sorts sites into size-rank buckets from Small (0–10 MW) to Mega campus (over 1,000 MW), but just over half of the sites (52%) are listed as Unknown. Rather than drop those sites or guess their size, we one-hot encoded the buckets and kept Unknown as its own category, giving 6 columns. Where it was available, we also used the reported power capacity, building square footage and land acreage. These range from a few megawatts to several thousand, so we applied a log1p transform to keep a handful of mega campuses from dominating, bringing the size group to 9 columns.

Power and cooling came from free-text fields that our cleaning notebook had already split into one binary flag per source: 14 for power, such as natural gas, grid or solar, and 6 for cooling, such as air, water or closed loop, for 20 in total. Community combined a single *pushback* flag with a one-hot encoding of how organized the resistance is, giving 6 columns. Media coverage came from counting FracTracker's cited source links and noting whether it had added any notes. To prevent leakage, we never read the contents of either, so the model can't pick up a company name. Electricity price came from outside FracTracker: the state's average 2025 industrial rate from EIA-861M. The last group, missing information, came from the data itself. So many fields were blank that we added 5 flags recording whether FracTracker had a value for square footage, acreage, project cost, building count and operator. Altogether, this gave 50 features for each of the 1,720 facilities.

We compared three classifiers, one linear and two nonlinear: logistic regression, a random forest, and k-nearest neighbors. Each makes a different assumption about how crypto and non-crypto facilities differ. Logistic regression assumes the difference can be captured by a weighted sum of features. The random forest assumes it lies in thresholds and combinations of features. KNN assumes crypto facilities resemble one another. Comparing them tells us something about the data as well as about which model scores best.

We first ran each model with simple starting settings, then tuned them with GridSearchCV, which tries every combination of settings from a list we provide and scores each one with cross-validation. Cross-validation splits the training data into five equal parts, called folds. The model is trained on four folds and scored on the fifth, and this repeats until every fold has been scored once. Averaging the five scores gives a more reliable estimate than a single split, and their spread shows how much the result depends on which sites happen to land in each fold. With only 35 crypto sites among 1,290 training rows, this mattered. We stratified the folds so each one holds about seven crypto sites. Without intervention, a fold could end up with very few crypto sites, or none, and its score would mean little. We used five folds instead of ten for the same reason: ten folds would leave only three or four crypto sites in each. For each model, we tuned the settings that control how closely it fits the training data: regularization strength for

logistic regression, tree depth and leaf size for the random forest, and the number of neighbors and vote weighting for KNN.

Evaluation

Our primary evaluation metric is the F1 score for the crypto class. We used it to guide our hyperparameters and to compare models. F1 combines two measures, recall and precision. Recall is the share of true crypto sites the model finds out of all the crypto samples. Precision is the share of sites the model labeled as crypto that are true crypto sites versus the total share of sites the model thought should be labeled as crypto. F1 combines the two into a single score that is high only when both are high, so a model cannot score well by flagging everything (high recall, low precision) or by flagging almost nothing (high precision, low recall). We chose F1 over accuracy because our data is so sparsely populated with labeled data. A model that never predicts crypto is 97.2% accurate on our test set without any feature consideration.

Model	**Family**	**CV F1**	**CV recall**	**CV precision**	**CV average precision**	**Test recall**	**Test precision**
Baseline (always "not crypto")	—	0.00	0.00	—	—	0.00	—
Logistic regression	Linear	0.31 ± 0.04	0.69	0.20	0.31	0.42	0.10
Random forest	Tree-based	**0.32 ± 0.05**	0.57	0.23	**0.42**	0.42	0.17
K-nearest neighbors	Instance-based	0.22 ± 0.07	0.20	0.30	0.08	0.08	0.14

Table #4 - Supervised Model Comparison

All three models beat the baseline, but they balance recall and precision differently. Logistic regression and the random forest have nearly the same F1 (0.31 and 0.32), but they get there in different ways. Logistic regression finds more of the crypto sites (recall 0.69), but only about 1 in 5 of the sites it flags is actually crypto (precision 0.20). The random forest finds fewer crypto sites (recall 0.57), but a slightly larger share of its flags are correct (precision 0.23). In practice, logistic regression would give a longer list of suspected crypto sites and the random forest a shorter list with fewer false alarms. We selected the random forest as our best model because it has the highest mean F1 and the highest average precision (0.42), although the F1 margin is small.

KNN performed the worst. KNN labels a site by looking at the most similar sites in the training data and taking a vote. Because 97% of sites are not crypto, most of any site's neighbors are not crypto either, so

the vote almost always comes out "not crypto." Grid search found that the only way around this was to use a single neighbor (k = 1), so KNN simply copies the label of the one most similar site. Even then, it found only about 1 in 5 crypto sites (recall 0.20). This suggests that crypto sites are not similar enough to one another, across our features, for a similarity-based method to pick them out.

As a final check, we scored each tuned model once on the held-out test set (430 sites, 12 crypto), which the models never saw during training or tuning. This shows how they perform on new data. Logistic regression and the random forest each found 5 of the 12 crypto sites, fewer than their cross-validation recall suggested. We see two reasons for this. First, the test set has so few crypto sites that each one changes recall by about 8 percentage points, so the result is noisy. Second, the cross-validation scores are slightly optimistic. Grid search tried many settings and kept the one that happened to score best on those folds.

In Depth Evaluation

The forest's feature importance ranks power capacity first by a wide margin, followed by building size, the "size unknown" flag and land size. Together, the size features account for 58% of the total importance. Permutation importance, which shuffles one feature at a time in the test set and measures how much F1 drops, instead ranks electricity price and the info_source flag highest. To settle the disagreement, we ran an ablation analysis: we removed one feature group at a time, retrained the forest, and measured the cross-validated F1. Removing the size features caused the largest drop, from 0.323 to 0.268, which agrees with tree importance. Permutation importance likely undervalues size because several features carry the same size information, so shuffling just one of them doesn't hurt much. Removing media coverage had the next largest effect (0.281). Price barely mattered (0.307), possibly because a state-wide average misses the site-specific rates large facilities negotiate. Removing the sparse power and cooling flags actually raised F1 to 0.355, so they may add more noise than signal. Most changes are close to the fold-to-fold variation (± 0.046), so we treat them as directions rather than firm conclusions.

We compared the cross-validated F1 for all 12 random forest settings in the grid search. Shallow trees underfit: every setting with a maximum depth of 3 scored between 0.22 and 0.24. The best setting (no depth limit, minimum leaf size of 5) scored 0.323 ± 0.046. With no depth limit and a minimum leaf size of 1, F1 dropped to 0.285 and varied far more across folds (± 0.158), because trees that grow to single-site leaves fit whichever few crypto sites land in each fold. The minimum leaf size mattered more for stability than for the average score.

At the default 0.5 threshold, the forest flags 29 test sites (recall 0.42, precision 0.17). Lowering the threshold to 0.2 finds 11 of the 12 crypto sites, but flags 156 sites at 7% precision. Raising it to 0.7 flags only 6 sites, half of them crypto, but recall falls to 0.25. A grid operator screening for possible new load would choose a low threshold, while anyone acting on individual flags would choose a high one. Training size also matters: validation F1 rose from 0.12 with 129 training rows to 0.32 with 1,032 and had not leveled off, so more labeled crypto sites would likely help more than further tuning.

Failure Analysis

The random forest made 31 mistakes on the test set: 24 false positives and 7 false negatives. To check whether they were due to chance, we retrained the forest with 20 different random seeds. 24 of the 31

mistakes were wrong with every seed, so most errors are consistent rather than random. We describe failures with four categories: systemic (a pattern the forest learned that is wrong for a whole group of sites), random (no clear cause; the mistake comes and goes with the seed), edge case (a rare kind of site with few similar training examples) and operational (a problem in our data pipeline rather than the model). Below we look closely at one example from three of these categories, using SHAP to show how each feature pushed the prediction toward or away from crypto. The "Seeds wrong" row in each table counts how many of the 20 retrained forests made the same mistake.

1. Lake Mariner Data Center: false positive, operational

Operator / state	TeraWulf/Fluidstack/Google/Core42, NY
Status / size	Expanding, Hyperscale, 500 MW, building size missing
Label vs forest	Not crypto vs 0.88 crypto
Seeds wrong	20 of 20
Evidence	Crypto tracker lists "Lake Mariner/ TerraWulf facility" 0.089 mi away; the match radius was 0.075 mi
Fix	Matching on operator name and a slightly wider radius would have caught this.

Table #5 - Failure Analysis - Lake Mariner

The forest was 88% sure this site was crypto, and it was right; our label was wrong. The crypto tracker lists the TeraWulf bitcoin facility 0.089 miles away so it did not match out is_crypto engineered feature being just outside the 0.075-mile match radius. TeraWulf has since signed agreements to host more than 200 MW of AI computing at the same campus (TeraWulf, 2025), so the site is both. Its large power capacity, press coverage and missing building size pushed it toward crypto, even though the model never saw the operator's name.

2. Amazon Madison Mega Site: false positive, systemic

Operator / state	Amazon, MS
Status / size	Operating, Hyperscale, 819 MW, building size missing
Label vs forest	Not crypto vs 0.68 crypto
Seeds wrong	20 of 20
Evidence	Building size is missing for 66% of crypto sites vs 39% of other sites; 20 of the 24 false positives are missing it
Fix	Fill a missing building size with the median for the site's size rank so a missing size stops pointing to crypto.

Table #6 - Failure Analysis - Amazon Madison

The forest rated this Amazon cloud campus 68% crypto. The largest push toward crypto came from its missing building size. Building size is missing for 66% of crypto sites but only 39% of other sites, so the forest learned that a missing size suggests crypto. That pattern is wrong for many large non-crypto campuses: 20 of the 24 false positives are also missing a building size.

3. Voltcore Bitcoin Mine: false negative, edge case

Operator / state	Not listed, AL
Status / size	Suspended, size Unknown, no power or building size
Label vs forest	Crypto vs 0.16 crypto
Seeds wrong	20 of 20
Evidence	Only 5 of the 891 sites with an unknown size rank are crypto
Fix	Another source for the site's power capacity, or county power-supply features, would give the forest evidence when the size is blank.

Table #7 - Failure Analysis - Voltcore

The forest gave this suspended bitcoin mine only a 16% chance of being crypto. FracTracker records no power capacity, building size or size rank for it, and the missing power capacity and unknown size rank pushed it away from crypto. Only 5 of the 891 sites with an unknown size rank are crypto, so the forest has almost no examples like it. An argument can be made that the site’s lack of similarity to other crypto sites was one of the causes for its suspension and brings a new perspective to the use of our models that were not originally considered in this project.

Discussion

A surprising outcome from the supervised model testing was the KNN’s lack of performance which signaled that all cryptocurrency data center sites are not similar in specific features but are similar across combinations of features. We originally hypothesized that the boundaries between crypto and non-crypto are clear, decisive and would be simple to extract from our base datasets. The importance of size features as accounting for >50% of feature importance was more expected and confirmed by the ablation analysis. Another finding was that the missing data values that we used as a way to understand the influence of data availability on crypto prediction actually cause systemic failures in labeling for our models.

The major challenges encountered for the supervised learning models include missing data, inconsistent labels, and increased need for data set cleaning to avoid data leakage. In many cases we chose to turn open string field data into categorical or binary columns to reduce the influence of data entry user bias, grammar, names, and repeated feature data between columns. Additionally, we had to combine multiple base datasets and overlay their features to boost the total number of data points available, as well as filling in features for the same sites that one dataset labeled but the second did not. The limited data effect on

hyperparameter tuning was limited by introducing stratification to ensure each fold and set of scores would utilize the same ratio of crypto/non-crypto rows.

With additional time and resources, we could find more data sources and reach out to wonders of private datasets to merge and create a large sample size that would enrich our model. Another use of time and resources would be to manually traverse the datasets and label/fill in missing data that is readily available or design an agent to scrape a subset of reputable sources and input that data.

Similar themes were discovered in the unsupervised learning model training. It was surprising to learn that our best performing model, the GMM, when looking at the Silhouette score only scored 0.30 portraying a weak cluster structure. The three clusters themselves were expected: high MW capacity, rural area with low electric costs, small MW capacity in highly populated areas that strain electrical grids with residential density, and a medium MW capacity with larger feature ranges.

One of the main challenges we encountered during the unsupervised section was limited feature availability. The dataset contained multiple attributes that were sparsely populated including *cooling source, cooling type, project cost,* and *power source*. Even attributes such as *facility size, acreage,* and *size rank (capacity)* only appeared in roughly 60% of the total data centers. When imputing median values or using the sparsely encoded features each model's performance dropped significantly and the resulting clusters were noisy and difficult to interpret. As a result, we decided to use a subset of the total data centers with high feature availability to examine any data centers archetypes that may appear in our dataset.

However, our approach does come with some limitations, potentially missing other subgroups that may have appeared if more data centers were included. Therefore, with more time and resources, we could include a larger number of data centers in the analysis, fill in missing rows, and uncover additional clusters or support existing structures on a larger portion of data centers.

Similarly to the supervised learning datasets, additional resources and time would go toward boosting the input datasets. For this type of modeling, we would also look to include additional features - other utility usage such as water, proximity or overlay of the country's public and private water servers, a general population density feature or proximity to commercial office buildings that may signal financial districts.

Ethical Considerations

From a privacy perspective, we experienced missing data or lack of data precision in locations and purpose of datacenters. There is a strong debate for considering data centers as critical infrastructures because they power cloud storage and processing of banking, industrial, and military operations. This had a significant impact on the availability of precise location and purpose data in our original datasets as facilities are safer from cybersecurity attacks if their information remains vague. As our project goal is to accurately predict centers, their purpose, and potential signals that the facility is migrating from one purpose to another, our models are open to potential attacks or abuse from malicious users as they could use our models to predict which data centers to target that would have the highest impact in cryptocurrency or AI sectors of our businesses and government operations. One way to address this would

be requiring end users to register or provide a client id/token from a handful of partners that would verify the legitimacy of the user account.

Additionally, the engineering features in this project rely heavily on the electrical grid power sources and generator supply which is a field that often sees technological advancements in areas such as renewable energy and power management. Without periodic updates and retraining as new power sources and generation is developed, the models could be abused to make a case against or for data centers conversions that are no longer relevant. For example, if our model features show a relationship between cryptocurrency centers being converted into AI centers and a significant increase in energy consumption and grid strain overall, but a specific facility starts to embrace more onsite renewable energy options, an end user may be able to use our model to bias a municipal decision to approve the project. Exploitation over time to bias municipalities and communities against expansion/rezoning is a strong driver to diligently watch model performance metrics such as drift/confidence and re-train the models when those values fall below a predefined threshold.

As these models are built on very limited labeled data, much of which is scraped or reported by community users, putting into question the bias of our source data. It is possible our original datasets and their feature markers for community pushback, resistance, and information sources hold bias against datacenter expansion and purpose migration. With limited data sourcing, the project is open to questions about bias and the project's choice to use potentially unrepresentative data and can be addressed over time as more data is available and the models are re-trained.

Finally, with the current state of the project there is a real question about releasing the models to the public without having serious consequences on business, brand damage, and overall trust between the public and data center companies. If the models were to be released, we would be responsible for providing a clear explanation of the model inputs/outputs and model performance metrics and warn end users to truly understand the infancy of the project versus the impact citing it may be. This is one tool for understanding data centers impact on communities and utilities and we would strongly urge that one tool does not tell the whole story and an end user should use other data points to make a final decision, such as the impact of a data center migration/expansion on the local job market, expansion of job skills, protections and other infrastructure improvements that may come with an approved project (road work, security protections, business tourism, and more).

Statement of Work

Cody Bauer	Matthew Chess	Lidiya Shutaya
Unsupervised learning model training and tuning, sensitivity analysis, data sourcing, and report methods and evaluation, related works	Supervised learning model training and tuning, ablation, failure analysis, data sourcing and report methods and evaluation	Data source research and exploration, data cleaning, feature engineering, ethical considerations, discussions and analysis, and report generation

References

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Tidy, Joe. "Ai Build out: Ai Gold Rush Draws Crypto Firms Away from Bitcoin." *BBC News*, BBC. 25 Aug. 2026, www.bbc.com/news/articles/c07ler8x9kpo.

Submission

- Link to GitHub repository (any files edited after submission = late submission)
    - Must be public or invite project coaches
- Link to Google Doc version of the written report with comments enabled: https://docs.google.com/document/d/1m3JfLKnmNOMxH6wEuN_WFDrnw7PUivFVlx6lkniTbB0/edit?usp=sharing

Appendix

Table #8. Final Features Table

Column_Name	Data Type	Description	Used in	Source
sizerank	categorical	Label based on range of capacity of a data center	Supervised	FracTracker
sizerank _numeric	numerical	Numeric representation to better suit model types	Unsupervised	Engineered from FracTracker
mw_high	numerical	Reported maximum electrical power capacity	Supervised	FracTracker
power_source	categorical	Community reported power source for the specific data center	Supervised	FracTracker
cooling_source	categorical	Reported cooling source for the data center	Supervised	FracTracker
cooling_type	categorical	Reported cooling type for	Supervised	FracTracker

		the data center		
facility_size_sqft	numeric	Reported data center size - buildings	Supervised, Unsupervised	FracTracker
property_size_acres	numeric	Reported data center size - land	Supervised, Unsupervised	FracTracker
is_crypto	binary	Binary label based on key words associated with cryptomining present in the facility name, purpose string, or included in column headings	Supervised	Engineered from FracTracker and Compute Atlas
community_pushback/pushback	categorical	Community Reported commentary on support or opposition to data center	Supervised	FracTracker
resistance_status	categorical	Level of opposition to data center operation	Supervised	FracTracker
project_cost	numerical	Reported project cost based on info_sources		FracTracker
info_source_1 - info_source_8	One-hot encoded	Where informational source or link was provided in each source column	Supervised	FracTracker
status	One-hot encoded	Current status of data center, operational, proposed, suspended or otherwise	Supervised	FracTracker
price/kWh	numerical	State reported price/kWh by customer type and month	Supervised	EIA-861M
energy_source (26 columns)	One-hot encoded	All utilized energy sources for generators by county and state. Differs from community reported power_source for specific data centers		Engineered from EIA-860
generator_count	numerical	Per county	Unsupervised	Engineered from EIA-860
generator_avg_time_to_full	numerical	Avg per county		Engineered from EIA-860
capacity_sum	numerical	Total per county	Unsupervised	Engineered from EIA-860
summer_Megawatthours	numerical	Total by general season by	Unsupervised	Engineered from

		State - Commercial only		EIA-861M
winter_Megawatthours	numerical	Total by general season by State- Commercial only		Engineered from EIA-861M
summer_cents/kWh.1/Industrial Price	numerical	Average by general season by State- Commercial only	Unsupervised	Engineered from EIA-861M
winter_cents/kWh.1/Industrial Price	numerical	Average by general season by State- Commercial only		Engineered from EIA-861M
has_sqft	binary	Whether this information was available in the original dataset	Supervised	FracTracker + Compute Atlas
has_acres	binary	Whether this information was available in the original dataset	Supervised	FracTracker + Compute Atlas
has_cost	binary	Whether this information was available in the original dataset	Supervised	FracTracker + Compute Atlas
has_buildings	binary	Whether this information was available in the original dataset	Supervised	FracTracker + Compute Atlas
has_operator	binary	Whether this information was available in the original dataset	Supervised	FracTracker + Compute Atlas